Restoration Ecology
THE JOURNAL OF THE SOCIETY FOR ECOLOGICAL RESTORATION

RESEARCH ARTICLE

Evaluating the efficacy of seasonal grazing and livestock exclusion as restoration tools for birds in riparian habitat of the Okanagan Valley, British Columbia, Canada

Timothy R. Forrester[1], David J. Green[1], René McKibbin[2], Christine A. Bishop[3,4]

Riparian habitat supports the highest density and diversity of songbirds in Western North America despite covering less than 1% of the land area. Widespread destruction and degradation of riparian habitat, especially by livestock grazing, has led to habitat restoration efforts. In 2000, restoration activities in the form of permanent and seasonal exclusion of livestock from riparian areas were initiated to improve habitat for the endangered Western Yellow-breasted Chat (*Icteria virens auricollis*) population, which is dependent on early successional shrub habitat for nesting, in the Okanagan Valley of British Columbia, Canada. We assessed the effectiveness of livestock exclusion by examining temporal changes in the abundance, richness, and breeding performance of birds in restoration and reference sites. The abundance of W. Yellow-breasted Chats significantly increased between 2002 and 2013 in areas where restoration activities occurred. However, restoration did not have significant effects on the abundance, richness, or breeding performance of other riparian birds at the restoration sites independent of temporal changes that occurred at reference sites. Our results provide evidence that limiting livestock grazing in temperate riparian areas can lead to recovery of endangered riparian songbirds that rely on early successional shrub habitat but may have limited effects on common species that are not strictly reliant on this habitat.

Key words: abundance, cattle, clutch size, nest success, productivity, Yellow-breasted Chat

Implications for Practice

- A combination of permanent and seasonal exclusion of livestock can lead to increased abundance of riparian birds that depend on early successional habitat for nesting such as Western Yellow-breasted Chats (*Icteria virens auricollis*).
- Common riparian birds may not show significant changes in abundance or breeding performance within the first decade of seasonal livestock exclusion in arid western valleys.
- Interpretation of temporal bird population responses to changes in livestock grazing regime should be validated by inclusion of reference as well as treatment data before and after change.
- Limiting grazing may not decrease brood parasitism by Brown-headed Cowbirds (*Molothrus ater*), which can negatively influence the breeding performance of riparian songbirds.

Introduction

Riparian habitat, the wetland interface between land and lakes or rivers (Chaney et al. 1990), supports the greatest diversity and highest densities of songbirds in Western North America (Knopf et al. 1988; Dobkin et al. 1998; Skagen et al. 1998). Riparian habitats are vulnerable to anthropogenic activities such as development, agriculture, river channelization, and livestock grazing (Patten 1998; Miller et al. 2003). Grazing can degrade riparian habitat by reducing shrub cover and eroding stream banks (Chaney et al. 1990; Fleischner 1994; Brown & McDonald 1995). As a consequence of human population growth, riparian habitat has been lost at a high rate in North America (Krueper 1993) and now comprises less than 1% of western land area (Chaney et al. 1990). Destruction and degradation of riparian habitat, as well as development of the surrounding landscape,

Author contributions: TRF, DJG, CAB conceived and designed the research; TRF, CAB, RM performed the research; TRF, DJG, CAB analyzed the data; DJG, CAB, RM contributed equipment and funding; TRF, DJG, CAB, RM wrote and edited the manuscript.

[1]Centre for Wildlife Ecology, Department of Biological Sciences, Simon Fraser University, 8888 University Drive, Burnaby, British Columbia V5A 1S6, Canada
[2]Canadian Wildlife Service, Environment and Climate Change Canada, 5421 Robertson Road, Delta, British Columbia V4K 3-N2, Canada
[3]Wildlife Research Division, Science and Technology Branch, Environment and Climate Change Canada, 5421 Robertson Road, Delta, British Columbia V4K 3-N2, Canada
[4]Address correspondence to C. A. Bishop, email cab.bishop@canada.ca

doi: 10.1111/rec.12495
Supporting information at:
http://onlinelibrary.wiley.com/doi/10.1111/rec.12495/suppinfo

often reduces the abundance and species richness of riparian bird communities (Lichstein et al. 2002; Tewksbury et al. 2002; Smith & Wachob 2006).

Recognizing the importance of riparian habitat, conservation practitioners have attempted to restore degraded habitats by excluding cattle (Nelson et al. 2011) and replanting native vegetation (Kus 1998), among other strategies. Restoration can improve habitat quality and increase the abundance and richness of riparian songbirds in previously degraded riparian habitats (Krueper et al. 2003; Gardali et al. 2006; Earnst et al. 2012). The effects of restoration activities on the breeding performance of songbirds are less clear. The overall daily nest survival of seven songbird species and aboveground artificial nests in riparian areas of Nevada was significantly lower in grazed areas than in areas that had not been grazed for 30 years (Ammon & Stacey 1997). Yet, the combined nest success of 12 bird species in riparian areas of Pennsylvania did not differ between grazed and control areas (Popotnik & Giuliano 2000), perhaps because fenced riparian corridors may also attract predators (Ratti & Reese 1988) or because predation can be more influential than vegetation structure on nest success (Harrison et al. 2011). Because habitat quality influences clutch size (Dhondt et al. 1992), habitat restoration has the potential to improve clutch size and therefore also the number of fledglings from nests, but no studies have yet extended their analysis of the impacts of livestock exclusion to include these reproductive parameters.

The Okanagan Valley of south central British Columbia, Canada, at the northern tip of the Great Basin, is a typical arid western valley where the riparian bird community has suffered as a result of habitat destruction and fragmentation. This area contains a mosaic of ecosystems that provide breeding habitat to over 200 species of birds (Cannings et al. 2010), one of the most irreplaceable avian communities in the country (Warman et al. 2004). A large number and diversity of these birds, including species at risk, use riparian habitat at some point in their life cycle (Cannings et al. 1987). Historically, nutrient rich floodplains that supported riparian forests and dense understories of shrubs followed the length of the Okanagan River. However, rapid human settlement and development led to the loss of 58% of black cottonwood (*Populus trichocarpa*) habitat and 92% of water birch (*Betula occidentalis*) habitat since the mid-1800s (Lea 2008). The remaining riparian habitat has been severely fragmented and degraded by urban and agricultural activities, road building, and grazing (Lea 2008). Riparian habitat loss and fragmentation has putatively been a major factor in population declines of some riparian songbird species in the Okanagan Valley including the endangered Western Yellow-breasted Chat (*Icteria virens auricollis*; Environment and Climate Change Canada 2016). Additionally, Brown-headed Cowbird (*Molothrus ater*) brood parasitism of some riparian birds in the Okanagan Valley has occurred at high rates (Ward & Smith 2000; Morgan et al. 2006), consistent with other western habitats where Brown-headed Cowbird abundance was highest in riparian areas that are fragmented (Donovan et al. 1997) and/or within 2 km of agriculture (Tewksbury et al. 1999).

Restoration activities in the form of permanent and seasonal exclusion of livestock from riparian areas were initiated in the Okanagan Valley in 2000 to restore habitat by increasing shrub cover for the W. Yellow-breasted Chat population. We collected data in 2001–2004 and 2012–2014 to examine temporal trends in the abundance, richness, and breeding performance (including parameters such as clutch size and number of fledglings which have been ignored by other restoration studies) of birds in remnant riparian habitat of the Okanagan Valley. We assessed the efficacy of restoration activities by comparing temporal trends in protected reference sites to restoration sites that have experienced livestock exclusion and natural succession for a decade. We predicted that restoration would lead to increases in abundance and richness independent of any temporal changes at reference sites. We also predicted that habitat improvements as a result of restoration would lead to increases in clutch size and/or the number of fledglings per nest of riparian songbirds. However, we also predicted that restoration would not improve daily nest survival or reduce brood parasitism by Brown-headed Cowbirds due to extensive habitat fragmentation and the presence of agricultural activities within the Okanagan Valley.

Methods

Study Design

This study took place during 2001–2004 and 2012–2014 in 10 riparian sites (Fig. 1), ranging in size from 1 to 180 ha (Table 1), in the southern part of the Okanagan Valley of British Columbia. We classified these study sites as "reference" or "restoration." Reference sites contained dense stands of black cottonwood and water birch forest patches, interspersed with thick patches of wild rose (*Rosa* sp.) and other shrubs. Reference sites gained a moderate degree of protection at varying times after the 1950s by being designated as provincial parks, national wildlife refuges, or private reserves and had not been subjected to livestock grazing since being protected. As of the late 1990s, restoration sites still contained patches of riparian forest but had highly degraded shrub layers due to livestock grazing. Starting in 2000, livestock were excluded from restoration sites year round or during the songbird breeding season in an attempt to restore the riparian habitat by allowing the understory shrub layer to regrow (Fig. 2). Fencing was also installed at these sites between 2001 and 2006 to completely exclude livestock from main riparian corridors near water.

Our reference sites did not represent pristine habitat quality but these sites had more protection from livestock before and throughout the decade of this study than restoration sites. Reference sites, just like our restoration sites, were embedded in a fragmented and heavily developed landscape, surrounded by agriculture and urbanization, and could have been negatively affected by these landscape-scale disturbances (Smith & Wachob 2006; Lee & Rotenberry 2015). However, results from reference sites were necessary for interpretation of any changes that occurred at restoration sites. For example, we assumed that if abundance increased at both reference and restoration sites,

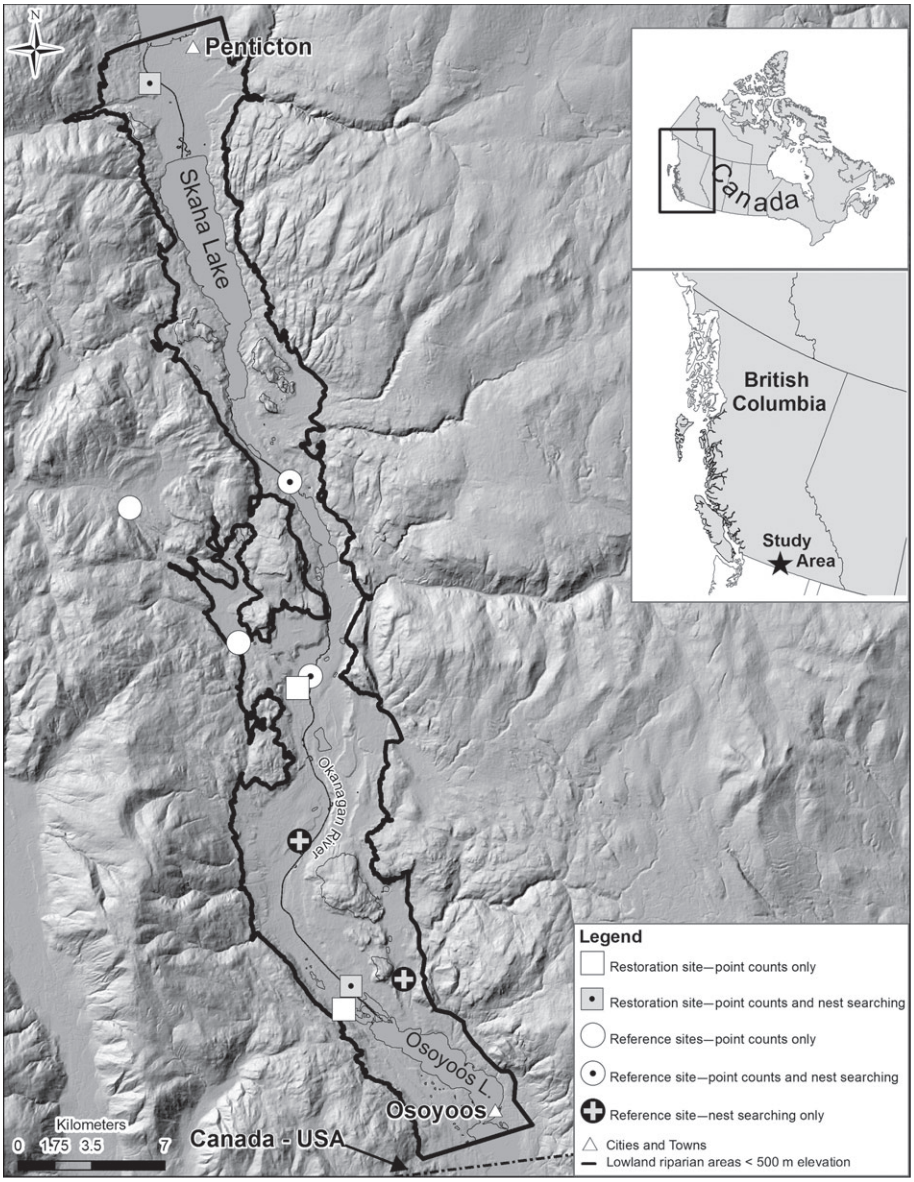

Figure 1. Riparian study area in the Okanagan Valley, British Columbia, Canada. The symbols indicate the category of each site (reference vs. restoration), as well as if abundance and richness were assessed at each site (using point counts), if breeding performance was monitored (using nest searching), or if both types of surveys were conducted.

we could not conclude that restoration influenced abundance. However if abundance increased at restoration sites and not at reference sites, we considered this evidence that restoration did affect abundance.

Sampling Methods

To assess the impact of habitat restoration measures on the abundance and richness of the riparian songbird community, we completed point counts in reference and restoration sites in 2001–2003 and 2012–2013. Due to difficulties in obtaining permission from landowners to access land, not all sites could be monitored every year (Table 1). We used an unlimited-radius point count protocol recommended by Dobkin et al. (1998). We did not calculate estimates of bird density using distance analysis because we estimated the distance of birds detected within 50 m of point count stations but not for birds beyond 50 m. The low number of point count stations required to cover this entire study area (Table 1) resulted in a relatively low number of detections for each species making it difficult to estimate density or detection curves. For these reasons, we defined abundance as the number of birds detected at each point count. Thus, our abundance values are indexes and not estimates of actual abundance.

We completed 10-minute point counts at the same 41 point count stations (in eight study sites) in 2001–2003 and

Table 1. Description of study sites in the Okanagan Valley, including site category (restoration vs. reference), size (ha), number of point count (PC) stations in each site, the years that abundance and richness were monitored using point counts, and the years the breeding performance was monitored using nest searching in each site. Restoration sites were treated with permanent and/or seasonal livestock exclusion after 2000. Reference sites were provincial parks, national wildlife refuges, or private reserves and were not grazed in recent history or during the study. These sites correspond with Figure 1, with Site 1 being the northernmost site and Site 10 being the southernmost site.

Site	*Category*	*Size (ha)*	*Number of PC Stations*	*Years Abundance and Richness Monitored*	*Years Breeding Performance Monitored*
1	Restoration	22	4	2002, 2012, 2013	2002–2004, 2012–2014
2	Reference	25	2	2001–2003, 2012, 2013	2003–2004, 2012, 2013
3	Reference	40	9	2001–2003, 2012, 2013	—
4	Reference	30	3	2001, 2003, 2012, 2013	—
5	Reference	5	1	2001–2003, 2012	2002–2004, 2012–2014
6	Restoration	1	1	2001–2003, 2012	—
7	Reference	15	0	—	2002–2004, 2012–2014
8	Reference	25	0	—	2002–2004
9	Restoration	180	15	2001–2003, 2012	2002–2004, 2012–2014
10	Restoration	55	6	2003, 2012	—

2012–2013. The number of point count stations at each site varied in relation to its size (Table 1). Survey effort varied by site and year; we visited reference sites an average of 5.3 ± 0.5 (SE) times (range = 4–6) in 2001–2003 and an average of 3.5 ± 0.5 times (range = 2–4) in 2012–2013. We visited restoration sites an average of 3.8 ± 1.3 times (range = 1–6) in 2001–2003 and an average of 2.5 ± 0.5 times (range = 2–4) in 2012–2013. We completed point counts once in early summer (26 May–15 June) and once in late summer (16 June–3 July), except for one restoration site (Site 1; Table 1) that we visited only once in late summer in 2001–2003. In 2001–2003, we visited some sites more than twice a year. In these cases, we only used data from the two point count surveys that occurred closest to the dates that we surveyed these sites in 2012–2013.

We conducted point counts only when there was no precipitation and winds were under 20 km/hour (Beaufort level three or lower). We initiated point counts at sunrise and concluded within 4 hours. Any differences in detection probability between 2001–2003 and 2012–2013 due to observer effects were minimized by having one of the observers from 2001–2003 train and validate the observer who conducted all of the surveys in 2012–2013. Simultaneous point counts conducted by these observers at the end of the training period were over 90% consistent for abundance and species richness. We assumed that any changes in vegetation would not affect detection probability.

To assess the impact of habitat restoration measures on the breeding performance of riparian songbirds, we searched for and monitored nests of five riparian songbird species in reference and restoration sites in May through mid-August of 2002–2004 and 2012–2014 (Table 1; Fig. 1). In 2012–2014, we surveyed five of the six sites that we surveyed in 2002–2004; access to the remaining reference site (Site 8; Table 1) was not available in 2012–2014. There were two reference sites in which we monitored breeding performance but did not also complete point counts. One (Site 7; Table 1) was composed of a few fragmented patches of riparian habitat along roadsides within a town where no point count stations were ever established because of the small size of the habitat patches. The other (Site 8; Table 1) was private land where we had permission to study W. Yellow-breasted Chats but not other species.

Our first focal species for monitoring breeding performance was the W. Yellow-breasted Chat (YBCH). Yellow-breasted Chats, including the western *auricollis* and the eastern *virens* subspecies, are large (22–27 g) songbirds that are widespread in riparian habitats throughout the United States and southern Canada (Eckerle & Thompson 2001) and have a global conservation status of least concern (BirdLife International 2012). The conservation status of the western subspecies has not been assessed though it is listed as a species of special concern in California (Comrack 2008) and declining population trends were reported in Washington and Oregon between 1968 and 2013 (Sauer et al. 2014). The Okanagan Valley is the northwest limit of the W. Yellow-breasted Chat breeding range. Elsewhere in Canada, a larger but disjunct population breeds in southeastern Alberta and southwestern Saskatchewan (Eckerle & Thompson 2001). The vast majority of W. Yellow-breasted Chats in British Columbia occur in the south Okanagan Valley and the nearby Similkameen Valley. Throughout the 1990s there were ≤20 pairs in British Columbia (Environment and Climate Change Canada 2016). This small population was designated as endangered by the Committee on the Status of Endangered Wildlife in Canada in 2000 and placed on the British Columbia provincial red list (which includes species classified as critically imperiled, imperiled, or vulnerable) in 2001. It was also listed as endangered under the Species at Risk Act in 2003 (Environment and Climate Change Canada 2016). We also monitored the breeding performance of four other species that are widespread in North America and common in the Okanagan Valley where they use riparian habitat for nesting. These were the Willow Flycatcher (WIFL; *Empidonax traillii*), Song Sparrow (SOSP; *Melospiza melodia*), Yellow Warbler (YEWA; *Setophaga petechia*), and Gray Catbird (GRCA; *Dumetella carolinensis*).

Following Martin and Geupel's (1993) protocol, we located nests using behavioral observations of adults and systematic searches. Nests were checked every 3 days, except around stage changing events when they were checked daily, to determine

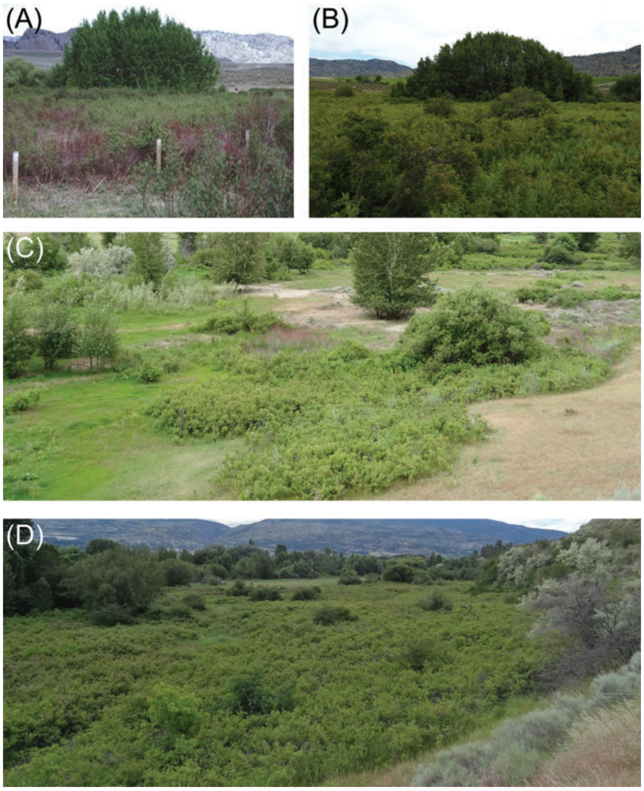

Figure 2. Riparian areas in the Okanagan Valley, British Columbia, Canada. (A) A restoration site in 2001 at the initiation of restoration activities, showing a fragmented and degraded shrub layer. (B) The same area as in (A) in 2016, 15 years after the initiation of restoration activities, showing a more continuous and higher shrub layer with occasional trees. (C) A riparian site in 2012 that was grazed throughout the study, showing fragmented patches of rose bushes (*Rosa* sp.) and a heavily degraded herbaceous layer. Because it was grazed throughout the study this site could not be classified as reference or restoration and was not used in this study. (D) A restoration site in 2012, 11 years after the initiation of restoration activities, showing a large, continuous patch of rose bushes complimented with occasional trees.

the date that incubation started, observed clutch size, occurrences of brood parasitism by Brown-headed Cowbirds, nest success (defined as fledging at least one fledgling, not including Brown-headed Cowbirds), and the number of young fledged. We aged nestlings based on feather development (Jongsomjit et al. 2007). We considered nests to have failed if the nest was obviously destroyed, if nestlings were missing more than 2 days before the anticipated fledging date, or if there was no evidence of fledglings or parental behavior near the nest within several weeks after the anticipated fledging date. We considered nests successful if they were empty within 2 days of the anticipated fledge date and if fledglings were observed, or if parental behavior indicated the presence of fledglings in the area.

Data Analysis

All analyses were conducted in R (R Core Team 2015). We utilized linear mixed effects models using package lme4 (Bates et al. 2015) to evaluate whether the total abundance and richness of all birds or the abundance of each of the five focal species and Brown-headed Cowbirds (per point count) varied with time period (2001–2003 vs. 2012–2013) or study site category (reference vs. restoration) while controlling for whether the survey was completed in early or late summer (survey time). We included site category, time period, and survey time as fixed effects. We also included interactions between study period and site category and between survey time and time period. Because we defined abundance as the number of birds detected per point count (rather than dividing by the number of point counts to get an average abundance per point count station), we accounted for repeated sampling at each point (in early and late summer of each year) by including point count station and year as random effects. These random effects ensured that our measures of abundance were not influenced by our different survey effort in 2001–2003 and 2012–2013.

We excluded Ring-billed Gull (*Larus delawarensis*) from the analysis because they were a “flyover” species that did not use riparian habitat, as well as large flocks (>20 individuals) of non-native species: European Starling (*Sturnus vulgaris*), House Sparrow (*Passer domesticus*), and Rock Pigeon (*Columbia livia*). Although one of the study sites was at least three times larger than any other (Site 9; Table 1), differences across time periods or between reference and restoration sites were not driven by data collected at this site; the results of analyses conducted excluding this site did not differ from those reported. Since we completed a different number of point counts in 2001–2003 and 2012–2013 (Table 2), we produced rarefaction curves to compare species richness between these time periods. Rarefaction curves estimate species richness as a function of the number of individuals detected, based on resampling from an existing dataset (Gotelli & Colwell 2001). We also analyzed species diversity using the Shannon–Wiener and Simpson’s diversity indices (Simpson 1949).

Next, we examined if different measures of breeding performance showed temporal trends or differed between reference and restoration sites while controlling for seasonal effects (incubation start date) and differences between years. We primarily focused on clutch size, the percentage of nests parasitized by Brown-headed Cowbirds, the number of fledglings from successful nests, and daily nest survival. We present and discuss the results of those four parameters (Table S1, Supporting Information). However, to allow comparison with other studies we also examined variation in overall productivity (number of young fledged from all nests, including nests that failed) and apparent nest success (Table S2).

We utilized linear mixed effects models to evaluate if clutch size, the percentage of parasitized nests, the number of fledglings, and apparent nest success of each of the five focal species varied with time period or with study site category. When evaluating clutch size, the number of fledglings, and nest success we included site category, time period, incubation start date, whether or not a Brown-headed Cowbird parasitized the nest, and an interaction between study period and site category as fixed effects. When evaluating the percentage of parasitized nests we included site category, time period, incubation start

Table 2. The number of point counts completed, number of individual birds detected, raw species richness (number of species detected), and diversity indices of the riparian songbird community in all sites as well as in reference and restoration sites of the Okanagan Valley in 2001–2003 and 2012–2013.

Time Period	*2001–2003*			*2012–2013*		
Site Category	*All*	*Reference*	*Restoration*	*All*	*Reference*	*Restoration*
Point counts completed	187	75	112	118	58	60
Individuals detected	3,976	1,407	2,569	3,025	1,431	1,594
Species richness	93	76	78	87	69	69
Shannon–Wiener index	3.71	3.62	3.57	3.65	3.52	3.53
Simpson's index	0.97	0.96	0.96	0.96	0.95	0.96

date, and an interaction between study period and site category as fixed effects. In those analyses, we included study site and year as random effects.

We calculated daily nest survival using the logistic-exposure method (Shaffer 2004). We used generalized linear models implemented in package mass (Venables & Ripley 2002) to evaluate whether daily nest survival varied between time period and study site category while controlling for differences between years. We included site category, time period, whether or not the nest was parasitized, year, and an interaction between study period and site category as fixed effects. For all analyses, visual inspection of residual plots did not reveal any obvious deviations from homoscedasticity or normality. We obtained chi-squared (χ^2) and p-values (p) by starting with a full model and sequentially dropped nonsignificant ($p > 0.05$) terms until only significant terms remained. Significance was assessed by comparing models with and without the term of interest using likelihood ratio tests (Pinheiro & Bates 2000). Values presented are means ± the 95% confidence interval.

Results

Abundance and Richness of the Riparian Bird Community

We detected significant increases to the total abundance and species richness (per point count) of the riparian bird community between 2001–2003 and 2012–2013 (Fig. 3A–B) but restoration did not have a detectable effect that was independent of the temporal changes that occurred at both reference and restoration sites. The total abundance of riparian birds detected per point count increased by 21% or 5.2 birds ($\chi^2_1 = 4.77$, $p = 0.03$). This pattern was observed at both reference and restoration sites (time period × site category interaction: $\chi^2_1 = 0.57, p = 0.44$; Fig. 3A). Total abundance was significantly higher at restoration sites than at reference sites throughout the study ($\chi^2_1 = 7.26$, $p < 0.01$). The abundances of all 105 species detected during this study are reported in Table S3. Species richness (per point count) also significantly increased at both reference and restoration sites but was significantly higher in restoration sites than in reference sites (time period: $\chi^2_1 = 5.98$, $p = 0.01$; site category: $\chi^2_1 = 5.14$, $p = 0.02$; interaction: $\chi^2_1 = 0.70$, $p = 0.40$; Fig. 3B). Diversity was high at both reference and restoration sites in both time periods but did not change over time or in response to restoration (Table 2). While the number of species detected at each point count increased over the last decade, we found no significant temporal changes to the overall rarefied species richness of riparian habitat in the Okanagan Valley; we detected 90.0 ± 3.0 species in 2001–2003 and 87.0 ± 0.0 species in 2012–2013. However, rarefied richness decreased at reference sites while it was lower and did not change over time at restoration sites (Fig. 3C).

Abundances of the Riparian Focal Species and Brown-headed Cowbirds

We found that the abundance of W. Yellow-breasted Chats significantly increased in restoration sites during the past decade (time period × site category interaction: $\chi^2_1 = 3.80$, $p = 0.05$); from 0.15 ± 0.16 to 1.12 ± 0.44 detections per point count (Fig. 4A). W. Yellow-breasted Chats were present in some reference sites but were rarely detected at the randomly selected point count stations. Restoration did not have a detectable effect on the abundance of the other riparian songbirds that was independent of temporal changes at both reference and restoration sites (time period × site category interaction for WIFL: $\chi^2_1 = 0.93$, $p = 0.34$; SOSP: $\chi^2_1 < 0.01$, $p = 0.95$; YEWA: $\chi^2_1 = 0.07$, $p = 0.79$; GRCA: $\chi^2_1 = 2.45$, $p = 0.12$; Fig. 4). Song Sparrows were significantly more abundant in reference sites than in restoration sites but their abundance did not significantly change over time (time period: $\chi^2_1 = 0.91$, $p = 0.34$; site category: $\chi^2_1 = 3.89$, $p = 0.05$; Fig. 4). There was a significant increase of Yellow Warbler abundance in both reference and restoration sites but their abundance did not significantly differ between these site categories (time period: $\chi^2_1 = 8.30$, $p < 0.01$; site category: $\chi^2_1 = 2.34$, $p = 0.13$; Fig. 4). There were no significant temporal changes in the abundance of Gray Catbirds or Willow Flycatchers (GRCA: $\chi^2_1 = 0.23$, $p = 0.63$; WIFL: $\chi^2_1 = 1.95$, $p = 0.16$) and the abundance of these species did not significantly differ between reference and restoration sites (GRCA: $\chi^2_1 = 1.02$, $p = 0.31$; WIFL: $\chi^2_1 = 1.95$, $p = 0.16$; Fig. 4). Brown-headed Cowbird abundance was significantly lower in reference sites than in restoration sites and significantly decreased in both site categories (time period × site category interaction: $\chi^2_1 = 3.84$, $p = 0.05$; Fig. 4).

Songbird Breeding Performance

We found that cattle exclusion did not lead to an improvement in the breeding performance of W. Yellow-breasted Chats at

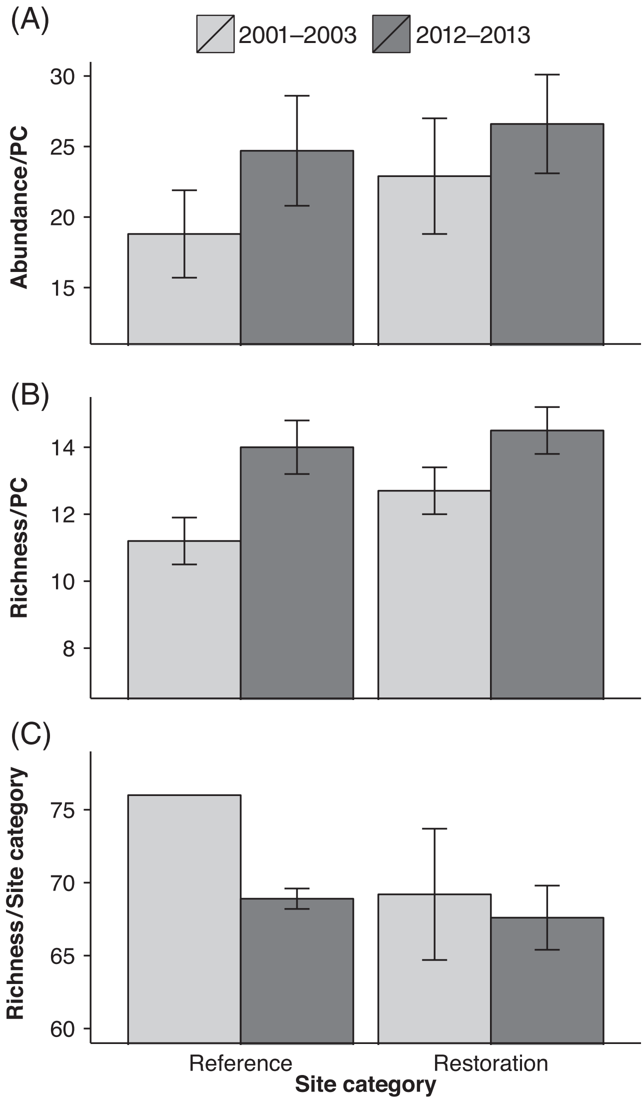

Figure 3. Abundance and richness ± 95% CI of all birds at reference and restoration sites in 2001–2003 and 2012–2013 in the Okanagan Valley: (A) average total abundance per point count (PC), (B) average species richness per point count, and (C) rarefied richness of reference and restoration sites. The rarefied richness of reference sites in 2001–2003 is the same as the raw species richness because the smallest number of individuals was detected in this category (Table 2), thus it has an error of zero.

restoration sites. Females at restoration sites produced significantly smaller clutches than females at reference sites (Table S1). However, the number of fledglings per successful nest, the percentage of parasitized nests, and the daily nest survival of W. Yellow-breasted Chats in restoration sites were the same as in reference sites and there was no evidence that their breeding performance significantly declined from 2002–2004 to 2012–2014 in restoration or reference sites (Table S1). Similarly, we found that cattle exclusion did not significantly improve the breeding performance of Willow Flycatchers, Song Sparrows, Yellow Warblers, or Gray Catbirds. Song Sparrows and Yellow Warblers in restoration sites had significantly higher rates of brood parasitism and significantly lower numbers of fledglings per successful nest than conspecifics in reference sites (Table S1). Song Sparrows in restoration sites also had significantly lower daily nest survival than in reference sites (Table S1). However, Willow Flycatchers and Gray Catbirds performed equally well at restoration and reference sites (Table S1). There was also no evidence to suggest that the breeding performance of any of those species declined over the last decade (Table S1).

Discussion

Western Yellow-breasted Chat abundance increased in riparian areas that were subjected to permanent and seasonal exclusion of livestock since 2000 in the Okanagan Valley. Reference sites, despite their close proximity to restoration sites, experienced virtually no change in W. Yellow-breasted Chat abundance while this endangered population increased dramatically in restoration sites. We consider this strong evidence that cattle exclusion improved the understory shrub habitat and encouraged increased abundance of W. Yellow-breasted Chats in restoration sites. However, counter to our predictions, the abundance and breeding performance of four other songbird species and the overall abundance and species richness of the riparian bird community did not increase under these conditions. A mix of seasonal and permanent livestock exclusion may have had limited impact on the wider riparian bird community because of a lower impact of seasonal versus permanent exclusion, the initial vegetative state of our restoration sites, the relative importance of early successional vegetation as nesting habitat, songbird sensitivity to cowbird parasitism, and because the abundance of riparian birds may be more strongly associated with landscape-scale characteristics than local-scale change.

Permanent removal of livestock from degraded riparian areas can be more beneficial to riparian bird communities than seasonal removal (Nelson et al. 2011). Complete removal of livestock from riparian habitat in the Great Basin led to increases in the abundance of Yellow Warblers, Willow Flycatchers, and Song Sparrows (Taylor & Littlefield 1986; Earnst et al. 2012). Although fences permanently protected most major riparian corridors in our restoration sites, livestock were still grazed in large areas of some of these sites outside of the breeding season. Seasonal removal of livestock from riparian sites in Colorado did not increase the abundance of Yellow Warblers, Willow Flycatchers, or Song Sparrows (Knopf et al. 1988). Our results provide evidence that a combination of seasonal and permanent removal of livestock from degraded riparian areas may be insufficient to enhance the abundance of all riparian songbirds and that complete exclusion of livestock and additional restoration actions may be required to enhance habitat quality and benefit most riparian songbirds.

The sites where cattle were excluded in this study were not completely devoid of riparian vegetation at the start of the study and even prior to restoration contained some patches of riparian forest and shrubs. Studies that found increased abundance of songbirds in response to restoration often commenced with

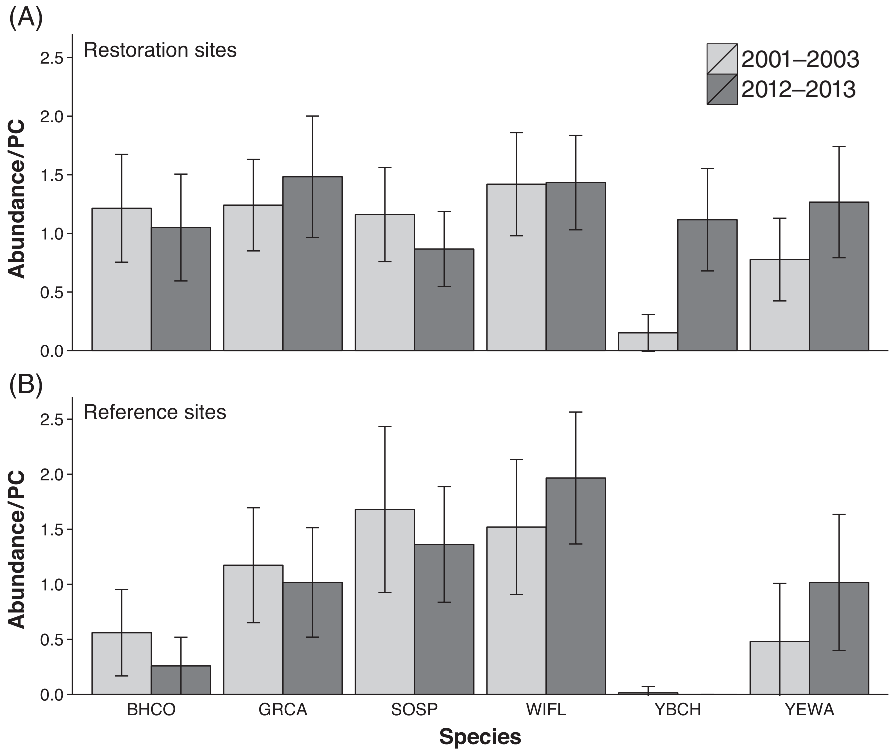

Figure 4. Average abundance ± 95% CI per point count (PC) of Brown-headed Cowbirds (BHCO), Gray Catbirds (GRCA), Song Sparrows (SOSP), Willow Flycatchers (WIFL), W. Yellow-breasted Chats (YBCH), and Yellow Warblers (YEWA) in the Okanagan Valley in 2001–2003 and 2012–2013 in (A) restoration sites and (B) reference sites.

sites that had almost no riparian vegetation relative to our sites (Farley et al. 1994; Dobkin et al. 1998; Gardali et al. 2006). Additionally, those studies were conducted at more southerly locations in the United States where the longer growing season may have allowed vegetative changes to occur faster than at our sites in Canada.

Cattle exclusion may have had a greater impact on W. Yellow-breasted Chats than other songbirds because in the Okanagan Valley W. Yellow-breasted Chats are heavily reliant on early successional vegetation for breeding (McKibbin & Bishop 2010). Because of this, Yellow-breasted Chats have been suggested to be good indicators of grazing pressure (Sedgwick & Knopf 1987). Yellow Warblers are often not negatively affected by controlled grazing (Knopf et al. 1988), possibly because they are not reliant on the understory shrub layer for nesting (Rich 2002). However, Willow Flycatchers and Song Sparrows are usually sensitive to grazing (Taylor & Littlefield 1986; Ammon & Stacey 1997), likely because they rely heavily on the shrub layer for nesting (Ammon & Stacey 1997; Rich 2002) and because they are especially vulnerable to brood parasitism (Knopf et al. 1988; Rogers et al. 1997). Persistent brood parasitism during this study may have prevented increases of these sensitive species. Indeed, not one parasitized Willow Flycatcher nest in this study was successful and Song Sparrows were also strongly affected (Forrester 2015). Although W. Yellow-breasted Chats in this study were also parasitized consistently, parasitism only slightly reduced their number of fledglings and did not affect their nest success (Forrester 2015), likely because their nestlings developed and fledged as rapidly as Brown-headed Cowbird nestlings (approximately 10 days; Bishop unpublished data) and were therefore not outcompeted.

The local restoration activities employed in a few areas of the Okanagan Valley may not have detectably increased the bird community in this study because riparian birds may be as dependent on the broader landscape within which remnant habitat is embedded as they are on the condition of their local habitat. The amount of riparian habitat left in a landscape can have a strong influence on both riparian bird abundance (Lee & Rotenberry 2015) and the effectiveness of restoration activities (Gardali & Holmes 2011). In an arid region of the southwestern United States, the abundance of riparian birds was more likely to be associated with landscape-scale characteristics than local-scale (Lee & Rotenberry 2015). In the Central Valley of California, riparian bird abundance increased more rapidly following restoration when restoration sites were located near large patches of existing riparian habitat (Gardali & Holmes 2011). Remnant riparian habitat in the Okanagan Valley is limited to only 8% of historic levels (Lea 2008) and represents a small proportion of the land area. The absence of community-wide responses to restoration in this study may therefore be a result of the general paucity of riparian habitat within this heavily modified landscape. Given the importance of the larger landscape, riparian restoration efforts should give priority to large habitat patches and account for a variety of sources of habitat degradation on multiple spatial scales (González et al. 2015).

Breeding Bird Surveys report declining population trends for Yellow Warblers, Willow Flycatchers, and Song Sparrows within British Columbia, Canada, and the western survey region during this study (2002–2013). In addition, populations of American Goldfinch (*Carduelis tristis*), American Robin (*Turdus migratorius*), Black-capped Chickadee (*Poecile atricapillus*), Northern Rough-winged Swallow (*Stelgidopteryx serripennis*), Tree Swallow (*Tachycineta bicolor*), and Veery (*Catharus fuscescens*) all declined in British Columbia or Canada during the same time (Sauer et al. 2014). However, only two Breeding Bird Survey routes occur in this study area and these routes only briefly pass through riparian habitat (Sauer et al. 2014). Thus, unlike Gardali et al. (2006) who used a formal comparison to Breeding Bird Survey results to evaluate the effects of restoration activities on a large scale in California, this study represents intensive surveys in a local area, comparing abundance and reproductive measures taken before restoration to the exact same survey points a decade later after restoration, with the added benefit of validating point count data between the observers who conducted the surveys 10 years apart. Our results suggest that populations of the same species that have declined in parts of North America have been relatively stable in riparian habitat of the Okanagan Valley over this same time period. Although our nest sample sizes were low in a few cases, which was partially a result of losing access to one reference site after 2004, our results from monitoring five songbird species also consistently showed that breeding performance did not decline over the last decade. This study therefore suggests that 10 years of natural habitat restoration in this valley may not have increased abundance or breeding performance of common riparian birds but had some local mitigating effect on potential declines in populations.

We conclude that restoration activities were successful in significantly increasing the endangered population of W. Yellow-breasted Chats but did not have a detectable impact on the abundance and richness of the bird community as a whole or on the breeding performance of riparian songbirds. Despite the large amount of agriculture and development in this unique area of Canada the remaining riparian habitat is capable of supporting an abundant, diverse, and productive community of birds.

Acknowledgments

This study was funded by the Federal Department of Environment and Climate Change Canada, Simon Fraser University, the Centre for Wildlife Ecology, and an NSERC Discovery grant. Simon Fraser University approved our Animal Care permit (#1039B-12). We thank C. Aikins, R. Armstrong, A.M. Bezener, D. Brown, O. Dyer, J. Emery, R. Hall, A. Huang, M. Lancaster, H. Lefort, T. Morgan, J. Ruddy, and J. St-Amand for assistance with data collection during 2012–2014. We also thank the En'owkin Centre, the Penticton Indian Band, the Osoyoos Indian Band, Environment and Climate Change Canada, the Nature Trust of British Columbia, the South Okanagan-Similkameen Conservation Program, Ducks Unlimited, the Nature Conservancy of Canada, the Ministry of Forests, Lands and Natural Resources Operations, Partners in Flight BC/Yukon, BC Parks, the British Columbia Ministry of Water, Land and Air Protection, the Federation of British Columbia Naturalists, the BC Important Bird Areas Program, the Canadian Nature Federation, Bird Studies Canada, Birdlife International, and the British Columbia Conservation Foundation for additional funding and logistical support. We also thank the private landowners who provided access to their properties and participated in riparian restoration efforts. We thank K. Mancuso for the 2016 habitat photograph. Insightful comments from K. Martin, R. C. Ydenberg, J. Isanhart, V. Amaral, and two anonymous reviewers improved the manuscript.

LITERATURE CITED

Ammon EM, Stacey PB (1997) Avian nest success in relation to past grazing regimes in a montane riparian system. Condor 99:7–13

Bates D, Mächler M, Bolker BM, Walker SC (2015) Fitting linear mixed-effects models using lme4. Journal of Statistical Software 67:1–48

BirdLife International (2012) *Icteria virens*. The IUCN red list of threatened species 2012: e.T22722057A39874194. 10.2305/IUCN.U.K. 2012-1.RLTS.T22722057A39874194.en (accessed 20 Oct 2016)

Brown JH, McDonald W (1995) Livestock grazing and conservation on southwestern rangelands. Conservation Biology 9:1644–1647

Cannings RA, Cannings RJ, Cannings SG (1987) Birds of the Okanagan Valley, British Columbia. University of British Columbia Press, Vancouver, Canada

Cannings RJ, Charlesworth C, Siddle C (2010) Checklist of the birds of the Okanagan Valley. South Okanagan Naturalists Club, Penticton, Canada

Chaney E, Elmore W, Platts WS (1990) Livestock grazing on western riparian areas. U.S. Environmental Protection Agency, Washington D.C.

Comrack LA (2008) Yellow-breasted Chat (*Icteria virens*). Pages 351–358. In: California bird species of special concern: a ranked assessment of species, subspecies, and distinct populations of birds of immediate conservation concern in California. Studies of Western Birds 1. Western Field Ornithologists/California Department of Fish and Game, Camarillo, California/Sacramento, California

Dhondt AA, Kempenaers B, Adriaensen F (1992) Density-dependent clutch size caused by habitat heterogeneity. Journal of Animal Ecology 61:643–648

Dobkin DS, Rich AC, Pyle WH (1998) Habitat and avifaunal recovery from livestock grazing in a riparian meadow system of the northwestern Great Basin. Conservation Biology 12:209–221

Donovan TM, Jones PW, Annand EM, Thompson FR III (1997) Variation in local-scale edge effects: mechanisms and landscape context. Ecology 78:2064–2075

Earnst SL, Dobkin DS, Ballard JA (2012) Changes in avian and plant communities of aspen woodlands over 12 years after livestock removal in the northwestern Great Basin. Conservation Biology 26:862–872

Eckerle KP, Thompson CF (2001) Yellow-breasted Chat (*Icteria virens*). In: Poole A, Gill F (eds) The birds of North America, No. 575. The Birds of North America Inc., Philadelphia, Pennsylvania

Environment and Climate Change Canada (2016) Recovery strategy for the Yellow-breasted Chat *auricollis* subspecies (*Icteria virens auricollis*) (Southern Mountain population) in Canada. Species at Risk Act Recovery Strategy Series. Ottawa. v + 31 pp

Farley GH, Ellis LM, Stuart JN, Scott NJ Jr (1994) Avian species richness in different-aged stands of riparian forest along the middle Rio Grande, New Mexico. Conservation Biology 8:1098–1108

Fleischner TL (1994) Ecological costs of livestock grazing in Western North America. Conservation Biology 8:629–644

Forrester TR (2015) Species richness, abundance and reproductive responses of riparian birds to habitat restoration in the Okanagan Valley. MS Thesis, Simon Fraser University, Burnaby, Canada

Gardali T, Holmes AL (2011) Maximizing benefits from riparian revegetation efforts: local- and landscape-level determinants of avian response. Environmental Management 48:28–37

Gardali T, Holmes AL, Small SL, Nur N, Geupel GR, Golet GH (2006) Abundance patterns of landbirds in restored and remnant riparian forest on the Sacramento River, California, U.S.A. Restoration Ecology 14:391–403

González E, Sher AA, Tabacchi E, Masip A, Poulin M (2015) Restoration of riparian vegetation: a global review of implementation and evaluation approaches in the international, peer-reviewed literature. Journal of Environmental Management 158:85–94

Gotelli NJ, Colwell RK (2001) Quantifying biodiversity: procedures and pitfalls in the measurement and comparison of species richness. Ecology Letters 4:379–391

Harrison ML, Mahony NA, Robinson P, Newbury A, Green DJ (2011) Nest-site selection and productivity of Vesper Sparrows breeding in grazed habitats. Journal of Field Ornithology 82:140–149

Jongsomjit D, Jones SL, Gardali T, Geupel GR, Gouse PJ (2007) A guide to nestling development and aging in altricial passerines. U.S. Department of Interior, Fish and Wildlife Service, Biological Technical Publication, FWS/BTP-R6008-2007, Washington D.C.

Knopf FL, Sedgwick JA, Cannon RW (1988) Guild structure of a riparian avifauna relative to seasonal cattle grazing. The Journal of Wildlife Management 52:280–290

Krueper DJ (1993) Effects of land use practices on western riparian ecosystems. Pages 321–330. In: Finch DM, Stangel PW (eds) Status and management of neotropical migratory birds. General Technical Report RM-229. United States Department of Agriculture, Forest Service, Washington D.C.

Krueper D, Bart J, Rich TD (2003) Response of vegetation and breeding birds to the removal of cattle on the San Pedro River, Arizona (U.S.A). Conservation Biology 17:607–615

Kus BE (1998) Use of restored riparian habitat by the endangered Least Bell's Vireo (*Vireo bellii pusillus*). Restoration Ecology 6:75–82

Lea T (2008) Historical (pre-settlement) ecosystems of the Okanagan Valley and Lower Similkameen Valley of British Columbia – pre-European contact to the present. Davidsonia 19:3–36

Lee M-B, Rotenberry JT (2015) Effects of land use on riparian birds in a semiarid region. Journal of Arid Environments 119:61–69

Lichstein JW, Simons TR, Franzreb KE (2002) Landscape effects on breeding songbird abundance in managed forests. Ecological Applications 12:836–857

Martin TE, Geupel GR (1993) Nest-monitoring plots: methods for locating nests and monitoring success. Journal of Field Ornithology 64:507–519

McKibbin R, Bishop CA (2010) Habitat characterization of breeding territories of the Western Yellow-breasted Chat in the South Okanagan, British Columbia, Canada. Northwestern Naturalist 91:145–156

Miller JR, Wiens JA, Hobbs NT, Theobald DM (2003) Effects of human settlement on bird communities in lowland riparian areas of Colorado (U.S.A). Ecological Applications 13:1041–1059

Morgan TC, Bishop CA, Bezener MA (2006) Temporal fluctuation in abundance of Brown-headed Cowbirds and their hosts in riparian habitat in the Okanagan Valley, British Columbia, Canada. Journal of Field Ornithology 77:444–451

Nelson KS, Gray EM, Evans JR (2011) Finding solutions for bird restoration and livestock management: comparing grazing exclusion levels. Ecological Applications 21:547–554

Patten DT (1998) Riparian ecosystems of semi-arid North America: diversity and human impacts. Wetlands 18:498–512

Coordinating Editor: John Isanhart

Pinheiro JC, Bates DM (2000) Mixed-effects models in S and S-PLUS. Springer-Verlag, Berlin, Germany

Popotnik GJ, Giuliano WM (2000) Response of birds to grazing of riparian zones. The Journal of Wildlife Management 64:976–982

R Core Team (2015) R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna. Austria. http://www.r-project.org (accessed 10 Jul 2014)

Ratti JT, Reese KP (1988) Preliminary test of the ecological trap hypothesis. The Journal of Wildlife Management 52:484–491

Rich TD (2002) Using breeding land birds in the assessment of western riparian systems. Wildlife Society Bulletin 30:1128–1139

Rogers CM, Taitt MJ, Smith JNM, Jongejan G (1997) Nest predation and cowbird parasitism create a demographic sink in wetland-breeding Song Sparrows. The Condor 99:622–633

Sauer JR, Hines JE, Fallon JE, Pardieck KL, Ziolkowski DJ Jr, Link WA (2014) The North American Breeding Bird Survey, Results and Analysis 1966–2013. Version 01.30.2015. USGS Patuxent Wildlife Research Center, Laurel, MD

Sedgwick JA, Knopf FL (1987) Breeding bird response to cattle grazing of a cottonwood bottomland. The Journal of Wildlife Management 51:230–237

Shaffer TL (2004) A unified approach to analyzing nest success. The Auk 121:526–540

Simpson EH (1949) Measurement of diversity. Nature 163:688

Skagen SK, Melcher CP, Howe WH, Knopf FL (1998) Comparative use of riparian corridors and oases by migrating birds in southeast Arizona. Conservation Biology 12:896–909

Smith CM, Wachob DG (2006) Trends associated with residential development in riparian breeding bird habitat along the Snake River in Jackson Hole, WY, U.S.A: implications for conservation planning. Biological Conservation 128:431–446

Taylor DM, Littlefield CD (1986) Willow Flycatcher and Yellow Warbler response to cattle grazing. American Birds 40:1169–1173

Tewksbury JJ, Martin TE, Hejl SJ, Redman TS, Wheeler FJ (1999) Cowbirds in a western valley: effects of landscape structure, vegetation, and host density. Studies in Avian Biology 18:23–33

Tewksbury JJ, Black AE, Nur N, Saab VA, Logan BD, Dobkin DS (2002) Effects of anthropogenic fragmentation and livestock grazing on western riparian bird communities. Studies in Avian Biology 25:158–202

Venables WN, Ripley BD (2002) Modern applied statistics with S. Springer, New York

Ward D, Smith JNM (2000) Interhabitat differences in parasitism frequency by Brown-headed Cowbirds in the Okanagan Valley, British Columbia. Pages 210–219. In: Smith JNM, Cook TL, Rothstein SI, Robinson SK, Sealy SG (eds) Ecology and management of cowbirds and their hosts. University of Texas Press, Austin, Texas

Warman LD, Forsyth DM, Sinclair ARE, Freemark K, Moore HD, Barrett TW, Pressey RL, White D (2004) Species distributions, surrogacy, and important conservation regions in Canada. Ecology Letters 7:374–379

Supporting Information

The following information may be found in the online version of this article:

Table S1. Primary measures of songbird breeding performance in the Okanagan Valley, British Columbia.
Table S2. Additional measures of songbird breeding performance in the Okanagan Valley, British Columbia.
Table S3. Abundances of all individual species detected in the Okanagan Valley in 2001–2003 and 2012–2013.

Received: 11 March, 2016; First decision: 17 May, 2016; Revised: 2 December, 2016; Accepted: 11 December, 2016; First published online: 19 January, 2017